

Below is a complete SRS draft based **only on the information you provided**. I have deliberately marked undecided implementation details as **TBD** rather than inventing requirements.

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

ServiceHub

Software Requirements Specification for the ServiceHub Platform

Version: 1.0

Project Name: ServiceHub

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Platforms: Web Application and Mobile Application

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the requirements for **ServiceHub**, a service marketplace platform designed to connect customers with trustworthy and reliable individual workers and businesses.

ServiceHub will be available through both a **web application** and a **mobile application**. The platform will allow customers to discover service providers, evaluate them based on information such as ratings, experience, portfolio, and pricing, and hire suitable providers.

The platform will support two primary methods of obtaining services:

1. **Direct hiring**, where a customer searches for a service provider and hires them.
2. **Job posting**, where a customer publishes a service request and workers or businesses apply with their proposed price and estimated completion time.

The purpose of this document is to provide a clear description of the system's functional and non-functional requirements and to serve as a reference for the design, development, testing, deployment, and future maintenance of the ServiceHub platform.

1.2 Scope

ServiceHub is intended to serve as a centralized marketplace for people who need services and professionals or businesses that provide those services.

The platform will support a broad range of service categories because the goal is to accommodate different types of workers rather than focusing on one specific profession.

Examples may include, but are not limited to:

- Home maintenance
- Construction
- Plumbing
- Electrical services
- Cleaning
- Automotive services
- Beauty and personal care
- Technology services
- Education and tutoring
- Photography
- Event services
- Professional services
- Other service-based occupations

The final service categories will be managed by the platform and may be expanded over time.

1.3 Intended Audience

This document is intended for:

- Software developers
- Mobile application developers
- Web developers
- Backend developers
- Database designers
- UI/UX designers
- System administrators
- Project supervisors
- Testers
- Project stakeholders
- Future maintenance teams

It may also be used as supporting documentation for academic project evaluation.

1.4 Definitions and Terminology

ServiceHub

The name of the service marketplace platform.

Customer

A user who searches for, hires, or requests services from workers or businesses.

Individual Worker

An individual professional who provides services through ServiceHub.

Business

A business or organization that provides services through ServiceHub.

Service Provider

A general term referring to either an individual worker or a business providing services.

Job

A service request created by a customer describing work that needs to be performed.

Application

A response submitted by a worker or business to a customer's posted job.

Portfolio

A collection of a worker's previous work, including photographs and certificates.

Rating

A numerical evaluation given by a customer to a worker after a service.

Review

Written feedback provided by a customer about a worker.

Administrator

A platform operator responsible for managing users, workers, businesses, services, reports, and other administrative functions.

2. Overall Description

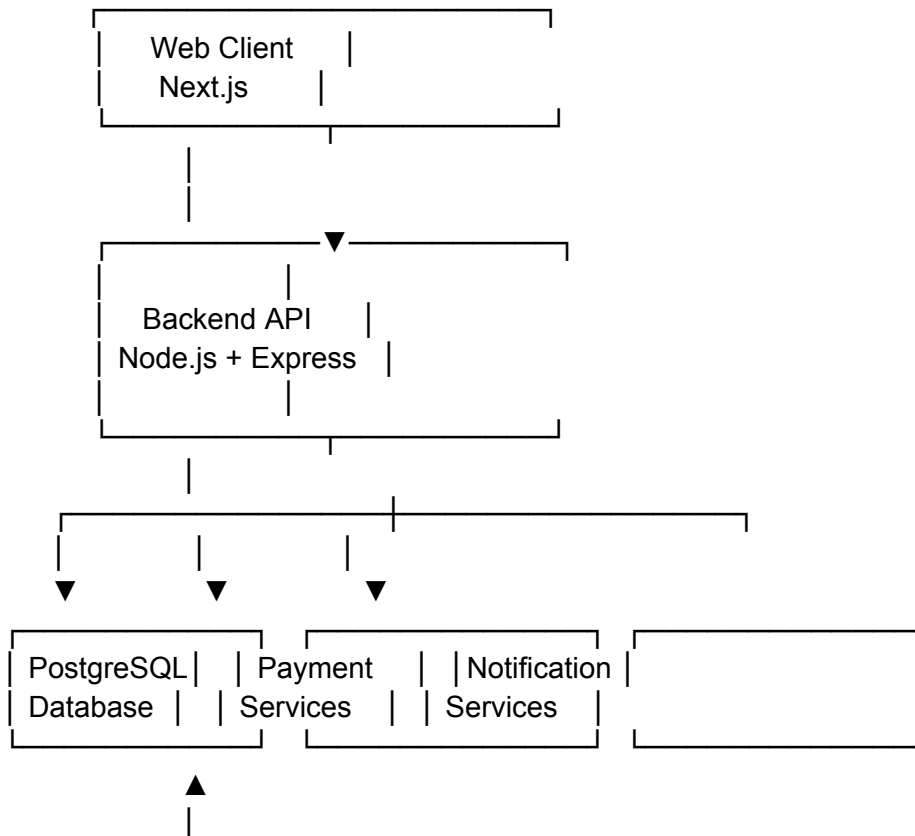
2.1 Product Perspective

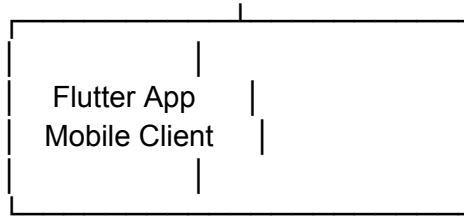
ServiceHub will operate as a multi-platform service marketplace.

The system will consist primarily of:

- Web frontend
- Mobile application
- Backend/API server
- Database
- Authentication system
- Payment integrations
- Notification services
- Administrative interface

The expected high-level structure is:





The web and mobile clients will communicate with the backend through APIs. The backend will manage business logic, authentication, data processing, and communication with external services.

2.2 Product Goals

The major goals of ServiceHub are to:

1. Connect customers with trustworthy service providers.
 2. Make service discovery easier.
 3. Provide customers with useful information before hiring.
 4. Allow workers to showcase their skills and experience.
 5. Provide workers with additional opportunities to find customers.
 6. Reduce reliance on informal intermediaries.
 7. Create a reputation-based service marketplace.
 8. Support both direct hiring and competitive job applications.
 9. Provide a platform where customers can evaluate workers through ratings and reviews.
 10. Create a scalable foundation for future services and features.
-

2.3 Target Users

The platform is intended for anyone who needs services and for professionals or businesses that provide them.

The primary user groups are:

Customers

People who need services and want to find and hire service providers.

Individual Workers

Independent professionals who provide services directly to customers.

Businesses

Organizations or businesses that provide one or more types of services.

Administrators

Authorized personnel responsible for managing and monitoring the platform.

3. System Scope

3.1 In-Scope Features for Version 1

The first version of ServiceHub will include:

- Customer registration
- Worker registration
- Business registration
- User authentication
- Phone verification
- Password reset
- Worker profiles
- Business profiles
- Service categories
- Service discovery
- Search
- Filtering
- Worker ratings
- Customer reviews
- Worker portfolios
- Certificates
- Direct hiring
- Customer job posting
- Worker job applications
- Proposed application pricing
- Estimated completion time
- Customer-worker communication

- Worker availability information
 - Emergency service requests
 - Notifications
 - Payment support
 - Cash payment
 - Telebirr payment
 - Chapa payment
 - Platform commission from workers
 - Administrative management
 - User reports
 - Web application
 - Mobile application
-

3.2 Features Not Included in the Initial Version

The following features have been identified as possible future features rather than confirmed Version 1 requirements:

- GPS-based location functionality
- Advanced location-based matching
- Multiple languages
- Offline functionality
- Advanced analytics
- AI recommendations
- Additional administrative roles
- Advanced verification badges
- Other advanced features not yet specified

These features may be considered in later versions.

4. Stakeholders and User Roles

4.1 Customer

The customer is a primary user of ServiceHub.

Customers can:

- Register.
 - Log in.
 - Search for services.
 - Search for workers.
 - Filter search results.
 - View worker profiles.
 - View worker portfolios.
 - View ratings and reviews.
 - View pricing information.
 - Contact workers.
 - Hire workers.
 - Post jobs.
 - Receive applications.
 - Compare applicants.
 - Select workers.
 - Request emergency services.
 - Make payments.
 - Rate workers.
 - Submit reviews.
 - Receive notifications.
-

4.2 Individual Worker

An individual worker is a professional who provides services independently.

Workers can:

- Register.
- Create a professional profile.
- Add services.
- Add skills.
- Add experience.
- Add pricing information.
- Upload portfolio photographs.
- Upload certificates.
- Display availability.
- Receive customer requests.
- Apply for posted jobs.
- Submit proposed prices.
- Provide estimated completion times.
- Communicate with customers.

- Complete jobs.
 - Receive customer ratings and reviews.
 - Receive notifications.
-

4.3 Business

A business represents an organization providing services.

Businesses will have functionality similar to service providers and will be able to:

- Register.
- Create a business profile.
- Describe their services.
- Provide service information.
- Display experience.
- Provide pricing information.
- Upload portfolio content.
- Upload certificates or relevant documentation.
- Receive customer requests.
- Apply for posted jobs.
- Communicate with customers.
- Receive ratings and reviews.
- Receive notifications.

The exact advanced management capabilities for businesses may be expanded in future versions.

4.4 Administrator

The administrator is responsible for operating and managing the platform.

The administrator will be able to manage:

- Users
- Workers
- Businesses
- Services
- Service categories

- Jobs
- Applications
- Reviews
- Reports
- Payments
- Platform-related information

Additional administrative roles may be introduced in future versions.

5. Product Functions

The major functions of ServiceHub are divided into the following modules:

1. Authentication and Account Management
 2. Customer Management
 3. Worker Management
 4. Business Management
 5. Service Management
 6. Search and Discovery
 7. Job Management
 8. Application Management
 9. Hiring Management
 10. Communication
 11. Availability
 12. Payment Management
 13. Rating and Review
 14. Notification Management
 15. Reporting
 16. Administration
-

6. Functional Requirements

6.1 Authentication and Account Management

FR-001: User Registration

The system shall allow users to create accounts.

The platform shall support different account roles including:

- Customer
- Individual Worker
- Business

The selected role shall determine the functions available to the user.

FR-002: Customer Phone Registration

The system shall require customers to register using a phone number.

The phone number shall be associated with the customer's account.

FR-003: Phone Verification

The system shall provide phone verification functionality.

A verification process shall be used to confirm the ownership of the phone number.

FR-004: Secure Login

The system shall allow registered users to authenticate securely.

Authentication shall use JWT-based authentication.

FR-005: Password Reset

The system shall provide a mechanism for users to reset their passwords.

FR-006: Role-Based Access

The system shall restrict functionality based on the user's role.

For example:

- Customers shall have access to customer functions.
 - Workers shall have access to worker functions.
 - Businesses shall have access to business functions.
 - Administrators shall have access to administrative functions.
-

6.2 Customer Profile Requirements

FR-007: Customer Profile

The system shall provide customer accounts and profiles.

Customer profile information may include:

- Name
- Phone number
- Account information
- Other required information

The exact profile fields may be refined during implementation.

6.3 Worker Profile Requirements

FR-008: Worker Profile Creation

The system shall allow workers to create professional profiles.

FR-009: Worker Information

A worker profile shall support information such as:

- Full name
- Phone number
- Email
- Profile photograph
- Bio
- Experience

- Skills
 - Services
 - Pricing information
 - Ratings
 - Reviews
 - Portfolio
 - Certificates
 - Availability
-

FR-010: Worker Portfolio

The system shall allow workers to upload photographs showing their previous work.

FR-011: Certificate Upload

The system shall allow workers to upload relevant certificates.

FR-012: Worker Experience

Workers shall be able to provide information about their professional experience.

FR-013: Worker Pricing

Workers shall be able to provide pricing/payment-rate information.

The exact pricing model and pricing rules are **TBD**.

FR-014: Worker Availability

Workers shall be able to provide their available working hours.

Customers shall be able to view the availability information.

6.4 Business Profile Requirements

FR-015: Business Profile

The system shall allow businesses to create service provider profiles.

FR-016: Business Services

Businesses shall be able to specify the services they provide.

FR-017: Business Information

Business profiles shall provide relevant information that allows customers to understand the business and its services.

The exact business verification and registration information is **TBD**.

6.5 Service Categories

FR-018: Service Categories

The system shall organize services into broad categories.

The platform is intended to support different types of workers rather than limiting itself to one profession.

Potential categories include:

- Home and maintenance services
- Construction services
- Automotive services
- Beauty and personal care
- Technology and digital services
- Education and tutoring
- Event-related services
- Transportation and delivery
- Professional services

- Other services

The final list of categories shall be configurable.

FR-019: Multiple Services

The system shall allow a worker or business to offer multiple services.

For example, one worker may provide:

- Plumbing
 - Water installation
 - Pipe repair
-

6.6 Search and Discovery

FR-020: Service Search

Customers shall be able to search for services.

FR-021: Worker Search

Customers shall be able to search for workers providing a particular service.

FR-022: Search Filters

The system shall provide filters for search results.

The confirmed filter includes:

- Rating

Additional filters may be implemented based on the final product design.

FR-023: Worker Profile Viewing

Customers shall be able to view worker profiles before hiring.

FR-024: Worker Evaluation Information

Customers shall be able to view relevant worker information including:

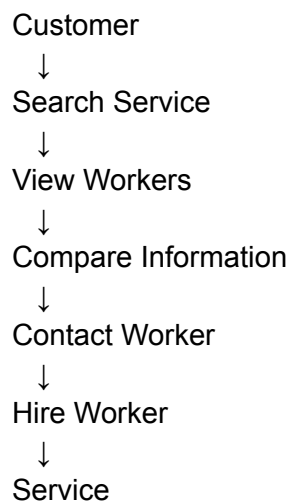
- Name
 - Bio
 - Experience
 - Photograph
 - Rating
 - Payment rate
 - Portfolio
 - Certificates
-

6.7 Direct Hiring

FR-025: Direct Hiring

The system shall allow customers to directly hire workers or businesses from the service discovery process.

The general workflow shall be:



↓
Payment
↓
Review

6.8 Job Posting

FR-026: Create Job

Customers shall be able to create and publish job requests.

FR-027: Job Information

A job posting shall support information such as:

- Job title
- Description
- Required service
- Budget
- Images
- Videos, where supported
- Deadline
- Location information
- Preferred experience
- Required skills

The exact fields may be refined during implementation.

FR-028: Job Visibility

Posted jobs shall be made available to suitable service providers according to the platform's matching and discovery mechanisms.

The exact matching algorithm is **TBD**.

6.9 Job Applications

FR-029: Worker Application

Workers shall be able to apply for jobs posted by customers.

FR-030: Application Price

A worker applying for a job shall be able to provide a proposed price.

FR-031: Completion Time

A worker applying for a job shall be able to provide an estimated completion time.

FR-032: Customer Application Review

Customers shall be able to view applications received for their jobs.

FR-033: Applicant Comparison

Customers shall be able to compare applicants based on available information such as:

- Proposed price
 - Estimated completion time
 - Worker profile
 - Experience
 - Ratings
 - Reviews
 - Portfolio
-

FR-034: Applicant Selection

Customers shall be able to select a worker or business from the applications received.

6.10 Communication

FR-035: Customer-Worker Communication

The system shall provide communication functionality between customers and service providers.

FR-036: Chat

Customers and workers shall be able to communicate using an in-platform chat system.

The detailed capabilities of the chat system are **TBD**.

6.11 Emergency Services

FR-037: Emergency Service Requests

The platform shall support emergency service requests.

Customers shall be able to identify a request as requiring urgent service.

The exact emergency notification and worker matching mechanism is **TBD**.

6.12 Payment

FR-038: Payment Methods

The platform shall support the following payment methods:

- Cash
- Telebirr
- Chapa

FR-039: Payment Recording

The system should maintain relevant payment information associated with services/jobs.

The exact payment architecture will depend on the final implementation.

FR-040: Platform Commission

ServiceHub shall generate revenue by taking a percentage of the worker's payment.

The exact percentage is **TBD**.

The commission rate should therefore be designed so it can be configured without major changes to the system.

FR-041: Escrow

The initial version of the platform shall **not implement escrow functionality**.

6.13 Ratings and Reviews

FR-042: Customer Rating

Customers shall be able to rate workers after receiving services.

FR-043: Customer Review

Customers shall be able to submit written reviews about workers.

FR-044: Worker Reputation

Worker ratings and reviews shall be associated with the worker's profile and contribute to the worker's reputation.

FR-045: Review Permissions

In Version 1, only customers shall review workers.

Worker-to-customer reviews are not part of the initial scope.

6.14 Notifications

FR-046: Notification System

The system shall notify users about relevant activities.

FR-047: Notification Channels

The system shall support:

- In-app notifications
 - Push notifications
 - Email notifications
-

FR-048: Relevant Notifications

Notifications may be generated for events such as:

- New job applications
- New messages
- Hiring events
- Job status changes
- Payment-related events
- Reviews
- Other important account activities

The final notification event list is **TBD**.

6.15 Reporting

FR-049: Report User/Service Provider

Users shall be able to report problematic users or service providers.

Potential report reasons include:

- Fraudulent behavior
- Misrepresentation
- Poor conduct
- Spam
- Other inappropriate behavior

FR-050: Administrative Report Management

Administrators shall be able to view and manage submitted reports.

6.16 Administration

FR-051: User Management

Administrators shall be able to manage registered users.

FR-052: Worker Management

Administrators shall be able to manage individual workers.

FR-053: Business Management

Administrators shall be able to manage registered businesses.

FR-054: Service Management

Administrators shall be able to manage service categories and service-related information.

FR-055: Job Management

Administrators shall be able to monitor and manage jobs posted on the platform.

FR-056: Application Management

Administrators shall be able to manage or monitor job applications when necessary.

FR-057: Review Management

Administrators shall be able to manage reviews and handle inappropriate or problematic reviews.

FR-058: Payment Management

Administrators shall be able to monitor payment-related information.

FR-059: Report Management

Administrators shall be able to review and manage user reports.

7. Non-Functional Requirements

7.1 Performance

NFR-001

The system should provide responsive interactions for normal user operations.

NFR-002

Search results should be returned within an acceptable amount of time under normal operating conditions.

NFR-003

The backend should be designed to support increasing numbers of users and service providers.

7.2 Security

NFR-004

The system shall use secure authentication.

NFR-005

JWT shall be used for authentication/session authorization.

NFR-006

Phone verification shall be implemented.

NFR-007

Password reset functionality shall be provided.

NFR-008

Users shall only be able to access functions permitted by their roles.

NFR-009

Sensitive user information shall be protected from unauthorized access.

7.3 Usability

NFR-010

The user interface should be simple and understandable.

NFR-011

The platform should allow customers to discover service providers without unnecessary complexity.

NFR-012

Worker profiles should clearly present important information such as services, experience, ratings, and pricing.

7.4 Scalability

NFR-013

The system architecture shall support future growth in:

- Users
 - Workers
 - Businesses
 - Services
 - Jobs
 - Applications
 - Transactions
-

7.5 Maintainability

NFR-014

The backend shall use a modular architecture.

NFR-015

The codebase shall follow appropriate software engineering practices.

NFR-016

The API should be structured in a way that allows both the web and mobile applications to consume the same backend services.

7.6 Compatibility

NFR-017

The platform shall be available through a web application.

NFR-018

The platform shall be available through a mobile application developed using Flutter.

NFR-019

The web application shall be developed using React/Next.js.

7.7 Availability and Reliability

NFR-020

The system should remain available to users during normal operating conditions.

NFR-021

The system should minimize data loss and service interruptions.

7.8 Localization

NFR-022

Version 1 shall use English as the primary application language.

The system may be designed so additional languages can be introduced in future versions.

8. User Requirements

8.1 Customer Requirements

A customer should be able to:

1. Register using a phone number.
 2. Verify their phone number.
 3. Log into the system.
 4. Search for services.
 5. Find suitable workers or businesses.
 6. Filter search results.
 7. View worker profiles.
 8. See worker experience.
 9. See worker ratings.
 10. See worker reviews.
 11. View worker portfolios.
 12. View certificates.
 13. View pricing information.
 14. Contact service providers.
 15. Hire a service provider directly.
 16. Post jobs.
 17. Receive applications.
 18. Compare applicants.
 19. Select an applicant.
 20. Request emergency services.
 21. Make payments.
 22. Rate workers.
 23. Write reviews.
 24. Receive notifications.
-

8.2 Worker Requirements

An individual worker should be able to:

1. Register.
 2. Verify their phone number.
 3. Log into the platform.
 4. Create a professional profile.
 5. Add services.
 6. Add skills.
 7. Add experience.
 8. Add pricing information.
 9. Add availability.
 10. Upload portfolio photographs.
 11. Upload certificates.
 12. Receive customer requests.
 13. Search for available jobs.
 14. Apply for jobs.
 15. Provide proposed prices.
 16. Provide estimated completion times.
 17. Communicate with customers.
 18. Receive job opportunities.
 19. Receive payments.
 20. Receive customer ratings and reviews.
 21. Receive notifications.
-

8.3 Business Requirements

A business should be able to:

1. Register.
2. Create a business profile.
3. Add services.
4. Provide service information.
5. Display experience and relevant information.
6. Upload portfolio materials.
7. Receive customer requests.
8. Apply for jobs.
9. Submit proposed prices.
10. Provide estimated completion times.

11. Communicate with customers.
 12. Receive customer ratings and reviews.
 13. Receive notifications.
-

8.4 Administrator Requirements

An administrator should be able to:

1. Log into the administrative system.
 2. Manage customers.
 3. Manage workers.
 4. Manage businesses.
 5. Manage service categories.
 6. Manage services.
 7. Manage jobs.
 8. Monitor applications.
 9. Manage reviews.
 10. Manage reports.
 11. Monitor payments.
 12. Manage platform-related information.
-

9. Use Cases

UC-01: Register Account

Actor: Customer / Worker / Business

Description:

A user creates a ServiceHub account.

Preconditions:

- User does not already have an account.
- User has a valid phone number.

Main Flow:

1. User opens registration.

2. User selects account type.
3. User enters required information.
4. System validates information.
5. System sends verification code.
6. User enters verification code.
7. System verifies the phone number.
8. System creates the account.
9. User can access the platform.

Postcondition:

A new verified account exists.

UC-02: Search for Worker

Actor: Customer

Main Flow:

1. Customer opens the search interface.
2. Customer enters the required service.
3. System searches available service providers.
4. System displays matching workers/businesses.
5. Customer applies filters if desired.
6. Customer selects a worker.
7. System displays the worker profile.

Postcondition:

Customer has access to the selected worker's information.

UC-03: Hire Worker Directly

Actor: Customer

Main Flow:

1. Customer searches for a service.
2. Customer views available workers.

3. Customer selects a worker.
 4. Customer reviews profile information.
 5. Customer contacts the worker if necessary.
 6. Customer hires the worker.
 7. Service is performed.
 8. Customer makes payment.
 9. Customer submits rating/review.
-

UC-04: Post Job

Actor: Customer

Main Flow:

1. Customer selects "Post Job."
 2. Customer enters job details.
 3. Customer specifies service requirements.
 4. Customer provides budget information.
 5. Customer provides deadline information.
 6. Customer provides other required details.
 7. Customer submits the job.
 8. System publishes the job.
 9. Suitable workers can view/apply.
-

UC-05: Apply for Job

Actor: Worker / Business

Main Flow:

1. Service provider views available jobs.
2. Service provider selects a suitable job.
3. Service provider reviews requirements.
4. Service provider enters proposed price.
5. Service provider enters estimated completion time.
6. Service provider submits application.
7. Customer receives notification.

UC-06: Select Worker

Actor: Customer

Main Flow:

1. Customer opens a posted job.
 2. Customer views applications.
 3. Customer compares applicants.
 4. Customer reviews profiles.
 5. Customer selects a service provider.
 6. System updates the job information.
 7. Selected provider is notified.
-

UC-07: Chat

Actors: Customer, Worker/Business

Main Flow:

1. User opens conversation.
 2. User sends a message.
 3. System delivers message.
 4. Recipient receives notification.
 5. Recipient replies.
 6. Conversation continues.
-

UC-08: Make Payment

Actor: Customer

Payment methods:

- Cash
- Telebirr

- Chapa

Main Flow:

1. Customer completes or arranges service.
 2. Customer selects payment method.
 3. Payment is completed according to the selected method.
 4. System records relevant payment information.
 5. Platform commission is calculated where applicable.
 6. Transaction information is associated with the relevant job.
-

UC-09: Rate Worker

Actor: Customer

Main Flow:

1. Customer completes a service.
 2. Customer opens the completed service.
 3. Customer provides a rating.
 4. Customer optionally provides written feedback.
 5. System stores the review.
 6. Worker's profile rating is updated.
-

UC-10: Report User

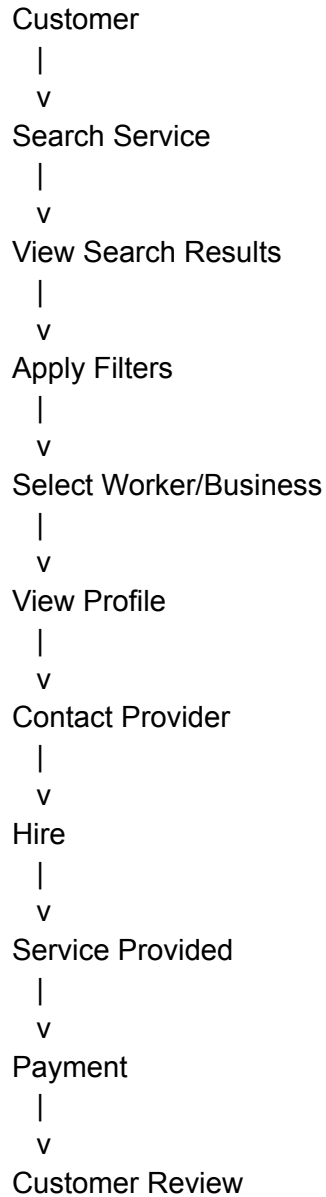
Actor: Customer / Worker / Business

Main Flow:

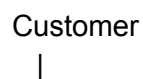
1. User selects report functionality.
 2. User identifies the problem.
 3. User submits report.
 4. System records report.
 5. Administrator reviews report.
 6. Administrator takes appropriate action.
-

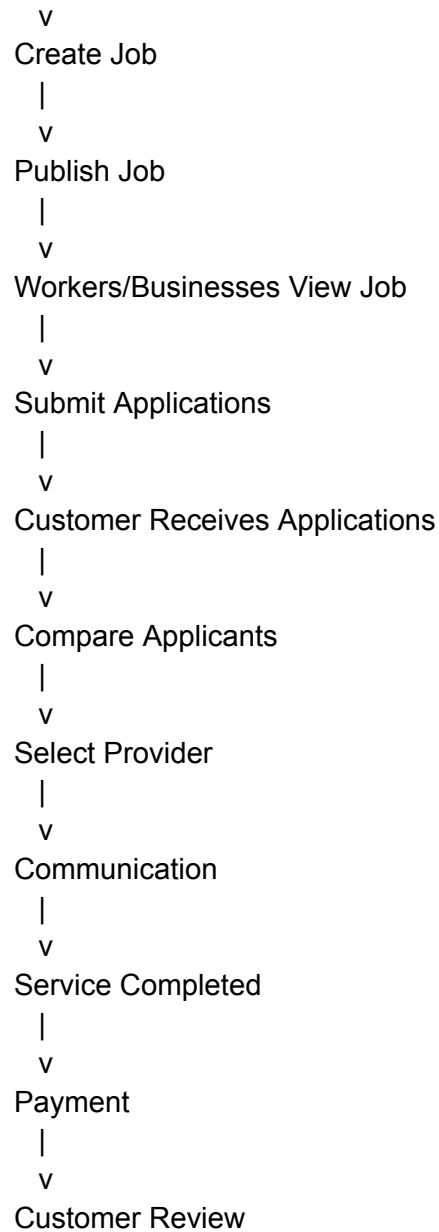
10. System Workflows

10.1 Direct Hiring Workflow

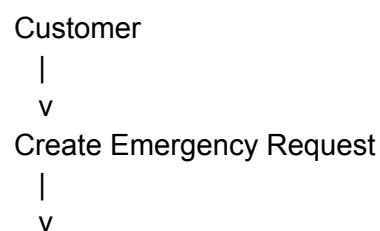


10.2 Job Marketplace Workflow





10.3 Emergency Service Workflow



System Identifies Suitable Service Providers

|

v

Available Providers Receive Notification

|

v

Provider Responds

|

v

Customer Selects Provider

|

v

Service Provided

|

v

Payment

|

v

Review

The exact matching and notification mechanism for emergency requests will be defined during detailed system design.

11. Data Requirements

The platform will require a relational database to manage users, services, jobs, applications, reviews, payments, and other system information.

PostgreSQL has been selected as the intended database technology.

11.1 User Entity

The User entity represents accounts registered on ServiceHub.

Possible attributes include:

- User ID
- Name

- Phone number
 - Email
 - Password information
 - Role
 - Account status
 - Created date
 - Updated date
-

11.2 Customer Entity

The Customer entity represents users who consume services.

Possible attributes include:

- Customer ID
 - User ID
 - Profile information
-

11.3 Worker Entity

The Worker entity represents individual service providers.

Possible attributes include:

- Worker ID
 - User ID
 - Full name
 - Bio
 - Experience
 - Profile photo
 - Payment rate
 - Availability
 - Rating information
-

11.4 Business Entity

The Business entity represents organizations providing services.

Possible attributes include:

- Business ID
 - User ID
 - Business name
 - Description
 - Contact information
 - Services
 - Experience
 - Portfolio
-

11.5 Service Entity

The Service entity represents services available through the platform.

Possible attributes include:

- Service ID
 - Service name
 - Description
 - Category ID
-

11.6 Service Category Entity

Possible attributes include:

- Category ID
 - Category name
 - Description
-

11.7 Portfolio Entity

The Portfolio entity stores worker portfolio information.

Possible attributes include:

- Portfolio ID
 - Worker ID
 - Title
 - Description
 - Image
 - Date
-

11.8 Certificate Entity

The Certificate entity stores certificates associated with workers.

Possible attributes include:

- Certificate ID
 - Worker ID
 - Certificate information
 - File reference
-

11.9 Job Entity

The Job entity represents a customer's service request.

Possible attributes include:

- Job ID
 - Customer ID
 - Title
 - Description
 - Budget
 - Deadline
 - Service
 - Status
 - Created date
-

11.10 Application Entity

The Application entity represents a worker/business application to a job.

Possible attributes include:

- Application ID
 - Job ID
 - Provider ID
 - Proposed price
 - Estimated completion time
 - Application status
 - Created date
-

11.11 Review Entity

The Review entity stores customer feedback.

Possible attributes include:

- Review ID
 - Customer ID
 - Worker/Business ID
 - Job ID
 - Rating
 - Comment
 - Created date
-

11.12 Payment Entity

The Payment entity represents payment-related information.

Possible attributes include:

- Payment ID
- Job ID
- Customer ID
- Provider ID

- Amount
 - Payment method
 - Payment status
 - Commission
 - Date
-

11.13 Notification Entity

Possible attributes include:

- Notification ID
 - User ID
 - Notification type
 - Message
 - Read status
 - Created date
-

11.14 Report Entity

Possible attributes include:

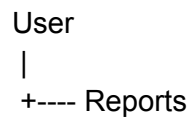
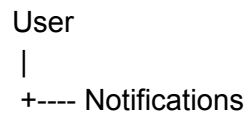
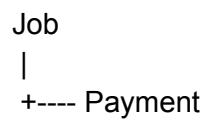
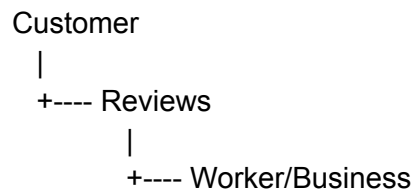
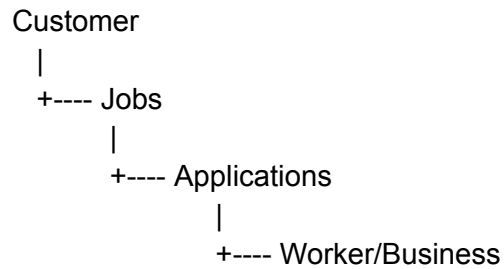
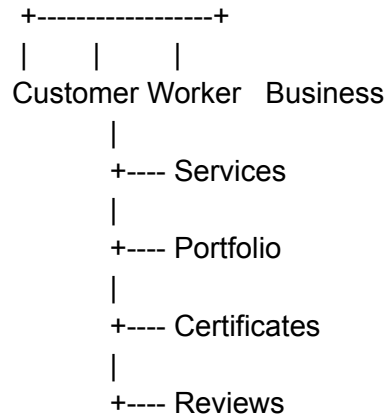
- Report ID
 - Reporting user
 - Reported user
 - Reason
 - Description
 - Status
 - Created date
-

12. Entity Relationships

A high-level relationship structure is:

User

|



The exact database relationships, cardinalities, normalization, and foreign-key structure will be defined during database design.

13. External Interface Requirements

13.1 Web Application

The web application will be developed using:

- React / Next.js
- Tailwind CSS

The web application will communicate with the backend through APIs.

13.2 Mobile Application

The mobile application will be developed using:

- Flutter

The mobile application will communicate with the same backend API used by the web application.

13.3 Backend

The backend will use:

- Node.js
- Express.js

The backend will provide APIs for:

- Authentication
- Users
- Workers
- Businesses
- Services
- Search
- Jobs
- Applications

- Hiring
 - Chat
 - Payments
 - Reviews
 - Notifications
 - Reports
 - Administration
-

13.4 Database

The intended database system is:

PostgreSQL

The database will store structured application data.

13.5 Payment Interfaces

The system will integrate with:

- Telebirr
- Chapa

Cash payment will also be supported.

The exact technical integration requirements for payment providers will be defined during implementation.

13.6 Notification Interfaces

The platform will provide:

- In-app notifications
- Push notifications
- Email notifications

The exact third-party notification providers are TBD.

14. System Architecture

ServiceHub will follow a layered client-server architecture.

14.1 Presentation Layer

The presentation layer consists of:

- Next.js web application
- Flutter mobile application

This layer provides interfaces for customers, workers, businesses, and administrators.

14.2 Application/API Layer

The backend will be implemented using Node.js and Express.js.

It will handle:

- Authentication
 - Authorization
 - Business logic
 - Data validation
 - Job management
 - Application management
 - Payment processing
 - Notifications
 - Reviews
 - Reporting
-

14.3 Data Layer

PostgreSQL will provide persistent data storage.

The backend will communicate with PostgreSQL through a database access layer.

The specific ORM/database library is currently **TBD**.

15. Security Requirements

Security is an important requirement because ServiceHub will manage personal information, communication, service information, and payment-related information.

15.1 Authentication

The system shall implement secure authentication using JWT.

15.2 Phone Verification

Users shall verify their phone numbers.

15.3 Password Security

Passwords shall not be stored in plain text.

They shall be securely hashed before being stored.

15.4 Authorization

The system shall enforce role-based authorization.

A customer shall not be able to access administrative functionality, and regular users shall not be able to perform administrator operations.

15.5 API Security

Backend APIs shall validate incoming requests and prevent unauthorized access.

15.6 Data Protection

Sensitive information shall be protected from unauthorized access.

16. Payment Requirements

ServiceHub will support three initial payment methods:

Cash

The customer and service provider may use cash for the service.

Telebirr

Customers will be able to use Telebirr where supported.

Chapa

Customers will be able to use Chapa-supported payment methods.

16.1 Platform Commission

ServiceHub will receive a percentage of worker earnings as a platform commission.

The commission percentage has not yet been determined.

Therefore:

Commission Rate = TBD

The payment architecture should allow this value to be configured in the future.

16.2 No Escrow in Version 1

The initial version will not provide an escrow system.

Escrow may be considered in future versions.

17. Notification Requirements

Notifications are an important part of ServiceHub because users need to know about new activities.

The system shall support:

In-App Notifications

Notifications displayed inside the application.

Push Notifications

Notifications delivered to mobile devices.

Email Notifications

Notifications delivered through registered email addresses where applicable.

18. Administration Requirements

The administrative system will provide centralized control over the platform.

18.1 User Management

Administrators shall be able to view and manage users.

18.2 Worker Management

Administrators shall be able to manage individual workers.

18.3 Business Management

Administrators shall be able to manage businesses.

18.4 Service Management

Administrators shall be able to manage service categories and available services.

18.5 Job Management

Administrators shall be able to monitor jobs posted through the platform.

18.6 Review Management

Administrators shall be able to manage inappropriate or problematic reviews.

18.7 Payment Management

Administrators shall be able to monitor payment-related information.

18.8 Report Management

Administrators shall be able to review reports submitted by users.

19. Business Rules

BR-001: Account Roles

Each account shall have an appropriate role.

The initial roles are:

- Customer
 - Individual Worker
 - Business
 - Administrator
-

BR-002: Customer Registration

Customers shall register using a phone number.

BR-003: Worker Services

A worker may provide multiple services.

BR-004: Business Services

A business may provide multiple services.

BR-005: Customer Reviews

Only customers can provide worker reviews in Version 1.

BR-006: Worker Reputation

Worker ratings and reviews contribute to their public reputation.

BR-007: Job Applications

Workers and businesses may apply to suitable customer jobs.

BR-008: Proposed Pricing

A worker or business applying to a job can provide a proposed price.

BR-009: Completion Time

A worker or business applying to a job can provide an estimated completion time.

BR-010: Commission

ServiceHub takes a percentage of worker earnings.

The exact percentage is TBD.

BR-011: Escrow

Escrow is not supported in Version 1.

BR-012: Language

English is the only supported language in Version 1.

20. Constraints

The development of ServiceHub is subject to the following constraints.

20.1 Technology Constraints

The intended technology stack includes:

- Next.js / React
 - Flutter
 - Node.js
 - Express.js
 - PostgreSQL
 - Tailwind CSS
-

20.2 Platform Constraints

The system must support both:

- Web
 - Mobile
-

20.3 Language Constraint

Version 1 will use English.

20.4 Payment Constraints

The initial payment options are:

- Cash
 - Telebirr
 - Chapa
-

20.5 Escrow Constraint

Escrow is outside the scope of Version 1.

21. Assumptions and Dependencies

21.1 Assumptions

The system assumes that:

- Users have access to a mobile phone.
 - Customers can access the internet.
 - Workers have sufficient information to create professional profiles.
 - Workers can provide accurate information about their services.
 - Customers will use ratings and reviews to evaluate service providers.
 - Payment providers will be available for integration.
 - Users will provide accurate information.
-

21.2 Dependencies

ServiceHub depends on:

- Internet connectivity
 - PostgreSQL database
 - Backend server
 - Payment services
 - Phone verification services
 - Email notification services
 - Push notification infrastructure
 - Web hosting
 - Mobile application infrastructure
-

22. Limitations

The first version of ServiceHub will have some limitations.

22.1 Geographic Location

Advanced GPS-based location functionality is not confirmed for Version 1.

22.2 Language

Only English will be supported initially.

22.3 Offline Operation

Offline functionality is planned for future versions and is not part of the initial requirements.

22.4 Analytics

Advanced analytics functionality is planned for future versions.

22.5 Verification

Advanced verification mechanisms and verification badges may be introduced in later versions.

22.6 Administrative Roles

Version 1 will use the basic administrator concept.

Additional specialized roles may be added later.

23. Future Enhancements

ServiceHub is intended to be expandable beyond the initial version.

Possible future features include:

23.1 Multiple Languages

Future versions may support:

- Amharic
 - Tigrinya
 - Afaan Oromo
 - Other languages
-

23.2 GPS and Location Services

Future versions may use GPS to:

- Find nearby workers.
 - Calculate distance.
 - Match customers with nearby providers.
 - Provide location-based services.
-

23.3 Advanced Analytics

The platform may later provide detailed analytics for:

- User growth
 - Worker performance
 - Customer activity
 - Popular services
 - Revenue
 - Job completion
 - Monthly active users
-

23.4 Additional Administrative Roles

Future versions may introduce:

- Super Administrator
 - Moderator
 - Finance Administrator
 - Customer Support Administrator
-

23.5 Advanced Verification

The platform may introduce:

- Verified Worker badges
 - Verified Business badges
 - Identity verification
 - Professional verification
-

23.6 Advanced Recommendation System

Future versions may recommend workers to customers based on:

- Service type
 - Rating
 - Previous performance
 - Location
 - Availability
 - Price
 - Customer preferences
-

23.7 Escrow

A future version may introduce escrow to improve payment security.

23.8 Additional Payment Methods

Additional local and international payment methods may be added as the platform expands.

24. Testing and Evaluation Requirements

Testing will be performed throughout development.

24.1 Unit Testing

Individual functions and components will be tested independently.

Examples include:

- Authentication functions
 - Search functions
 - Rating calculations
 - Job creation
 - Application submission
 - Payment calculations
-

24.2 Integration Testing

Integration testing will verify communication between different system components.

Examples:

- Frontend → Backend
 - Backend → Database
 - Backend → Payment services
 - Backend → Notification services
-

24.3 API Testing

The REST APIs will be tested to ensure:

- Correct requests are accepted.
 - Invalid requests are rejected.
 - Authentication works correctly.
 - Authorization works correctly.
 - Correct responses are returned.
-

24.4 UI Testing

The user interfaces will be tested to ensure:

- Navigation works.
 - Forms work correctly.
 - Search works correctly.
 - Profiles display properly.
 - Job posting works.
 - Applications display correctly.
 - Mobile interfaces work correctly.
-

24.5 Security Testing

Security testing will verify:

- Authentication.
 - Authorization.
 - Password protection.
 - JWT handling.
 - Input validation.
 - Unauthorized access prevention.
-

24.6 User Acceptance Testing

Potential users will evaluate whether ServiceHub:

- Makes service discovery easier.
- Provides sufficient worker information.
- Makes hiring understandable.
- Makes job posting easy.

- Makes applications easy to compare.
 - Provides a satisfactory user experience.
-

25. Success Criteria

The success of ServiceHub will be evaluated using several indicators.

25.1 Registered Users

The number of registered users will be monitored.

This includes:

- Customers
 - Individual workers
 - Businesses
-

25.2 Customer Satisfaction

Customer satisfaction will be used as an important measure of the platform's success.

Feedback may be collected through:

- Reviews
 - Ratings
 - Surveys
 - Other feedback mechanisms
-

25.3 Monthly Active Users

The platform will track monthly active users as an indicator of ongoing engagement.

25.4 Service Activity

The number of:

- Jobs posted
- Applications submitted
- Workers hired
- Completed services

may also be used to evaluate platform activity.

26. Proposed High-Level User Interface

The platform will have different interfaces depending on the user's role.

26.1 Landing Page

The landing page should provide:

- ServiceHub branding
 - Service search
 - Service categories
 - Login
 - Registration
 - Introduction to the platform
-

26.2 Customer Dashboard

The customer dashboard may contain:

- Search services
 - Recommended/discovered workers
 - Posted jobs
 - Applications
 - Active jobs
 - Completed jobs
 - Messages
 - Notifications
 - Profile
-

26.3 Worker Dashboard

The worker dashboard may contain:

- Profile
 - Services
 - Portfolio
 - Certificates
 - Availability
 - Available jobs
 - Applications
 - Active jobs
 - Messages
 - Notifications
 - Reviews
-

26.4 Business Dashboard

The business dashboard may contain:

- Business profile
 - Services
 - Portfolio
 - Available jobs
 - Applications
 - Active jobs
 - Messages
 - Notifications
 - Reviews
-

26.5 Worker Profile Page

The public worker profile should display:

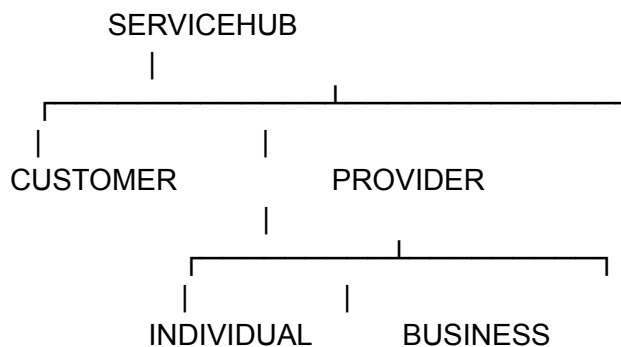
- Profile photo
- Name
- Bio
- Experience
- Services

- Pricing information
- Ratings
- Reviews
- Portfolio
- Certificates
- Availability

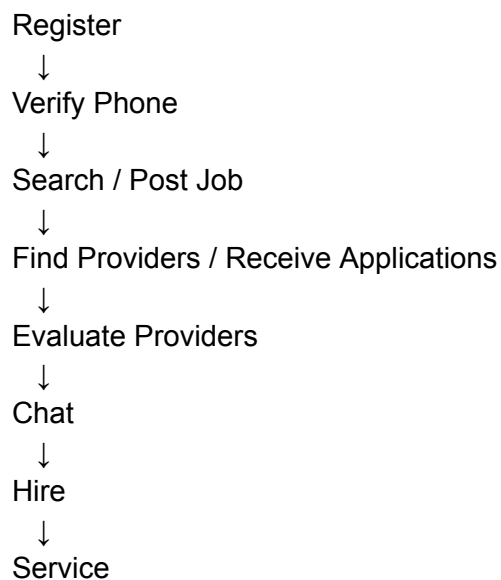
This page is an important component of the platform's trust model because customers should be able to evaluate providers before hiring.

27. Detailed Platform Interaction Model

The overall ServiceHub experience can be summarized as follows:



Customer path:



↓
Payment
↓
Review

Worker path:

Register
↓
Verify Account
↓
Create Profile
↓
Add Services
↓
Add Experience
↓
Add Portfolio
↓
Set Pricing/Availability
↓
Receive Requests / Find Jobs
↓
Apply
↓
Get Hired
↓
Provide Service
↓
Receive Payment
↓
Build Reputation

Business path:

Register
↓
Create Business Profile
↓
Add Services
↓
Add Portfolio/Information
↓

Receive Requests / Find Jobs



Apply



Get Hired



Provide Service



Receive Payment



Build Reputation

28. Trust and Reputation Model

Trust is one of the central concepts of ServiceHub.

Traditional service discovery often requires customers to depend on recommendations from friends, family, or informal intermediaries.

ServiceHub attempts to create a digital reputation system by allowing customers to evaluate service providers after receiving services.

A provider's profile can therefore contain:

Service Provider



+— Experience



+— Services



+— Pricing



+— Portfolio



+— Certificates



+— Completed Work



+— Customer Ratings



+— Customer Reviews

This information gives customers multiple factors to consider before selecting a provider rather than relying solely on personal recommendations.

29. Dual Marketplace Model

One of the key features of ServiceHub is that it does not rely on a single hiring model.

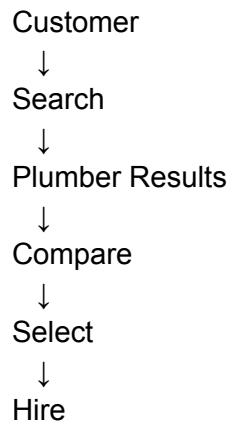
Model 1: Search-Based Hiring

The customer knows what service is needed.

Example:

"I need a plumber."

The customer searches for plumbers and evaluates available providers.



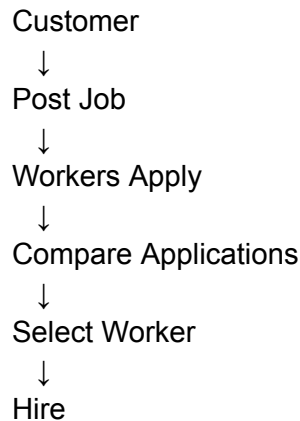
Model 2: Job-Based Hiring

The customer knows what needs to be done but may not know which provider to choose.

Example:

"I need someone to install electrical wiring in my house."

The customer creates a job.



This dual approach allows ServiceHub to serve different customer needs.

30. Requirements Traceability Overview

The major objectives can be traced to platform features.

Objective	Supporting Feature
Connect customers and providers	Service marketplace
Improve trust	Ratings, reviews, portfolios
Help customers find workers	Search and filters
Help workers find customers	Job marketplace
Allow customers to compare providers	Profiles, ratings, prices
Provide professional visibility	Worker/business profiles
Improve communication	In-platform chat
Support service transactions	Payment functionality
Handle problematic users	Reporting
Maintain platform quality	Administration
Measure platform growth	User and activity metrics

31. Technology Stack

The currently proposed technology stack is:

Frontend — Web

React / Next.js

Next.js will be used to build the web interface.

Styling

Tailwind CSS

Tailwind CSS will be used to implement the user interface styling.

Mobile

Flutter

Flutter will be used to build the mobile application.

Backend

Node.js + Express.js

Node.js and Express.js will provide the backend server and REST API.

Database

PostgreSQL

PostgreSQL will be used for persistent relational data storage.

Authentication

JWT

JSON Web Tokens will be used for authentication and authorization.

Payment

The initial payment ecosystem will include:

- Telebirr
 - Chapa
 - Cash
-

32. API Requirements

The backend should expose REST APIs for the major platform functions.

Potential API groups include:

/api/auth
/api/users
/api/customers
/api/workers
/api/businesses
/api/services
/api/categories
/api/search
/api/jobs
/api/applications
/api/hiring
/api/messages
/api/payments
/api/reviews
/api/notifications

/api/reports
/api/admin

The exact endpoint naming and API versioning strategy will be determined during backend development.

33. Data Integrity Requirements

The system shall maintain consistency between related entities.

For example:

- A job must belong to a customer.
- An application must belong to a job.
- An application must belong to a worker or business.
- A review must be associated with the appropriate service relationship.
- A payment must be associated with the relevant service/job.
- A portfolio item must belong to a service provider.
- Notifications must belong to users.

The database should use appropriate constraints and relationships to prevent invalid data.

34. Error Handling

The system should provide meaningful error messages when operations fail.

Examples include:

Authentication Error

Invalid phone number or password.

Validation Error

Required information is missing.

Job Error

Unable to create job. Please try again.

Payment Error

Payment could not be completed.

Network Error

Unable to connect to the server. Please try again.

Error messages should be understandable to users without exposing sensitive technical information.

35. Privacy Considerations

ServiceHub will handle user information including:

- Names
- Phone numbers
- Email addresses
- Profile information
- Portfolio information
- Reviews
- Payment-related information
- Messages

The system should therefore ensure that private information is only accessible to authorized parties.

Public profiles should expose only information intended to be visible to other users.

36. Scalability Considerations

The platform is expected to grow over time.

The architecture should therefore be capable of supporting increasing numbers of:

- Customers
- Workers

- Businesses
- Service categories
- Jobs
- Applications
- Reviews
- Messages
- Transactions

The use of a dedicated backend API and PostgreSQL database provides a foundation for future scaling.

37. Version 1 Feature Summary

The first version of ServiceHub will focus on the core marketplace experience.

Account System

- Customer accounts
- Individual worker accounts
- Business accounts
- Authentication
- Phone verification
- Password reset

Service Discovery

- Service categories
- Search
- Filtering
- Provider profiles

Provider Information

- Bio
- Experience
- Services
- Pricing
- Portfolio
- Certificates
- Ratings
- Reviews

- Availability

Hiring

- Direct hiring
- Job posting
- Applications
- Proposed prices
- Estimated completion times

Communication

- Customer-provider chat

Transactions

- Cash
- Telebirr
- Chapa
- Worker commission

Trust

- Ratings
- Reviews
- Portfolio
- Certificates
- Reporting

Administration

- User management
- Worker management
- Business management
- Service management
- Job management
- Review management
- Payment management
- Report management

Notifications

- Push
- Email

- In-app

Platforms

- Web
 - Mobile
-

38. Features Explicitly Reserved for Future Versions

The following should not be treated as mandatory Version 1 requirements:

- GPS-based location services
- Advanced geographic matching
- Offline support
- Multiple languages
- Advanced analytics
- AI-based recommendations
- Additional admin roles
- Advanced verification badges
- Escrow
- Other advanced functionality

This separation is important because it prevents future ideas from unnecessarily expanding the Version 1 development scope.

39. Acceptance Criteria

The initial ServiceHub implementation can be considered functionally successful when the following core scenarios work correctly.

Customer Scenario

A customer must be able to:

1. Register using a phone number.

2. Verify the account.
 3. Log in.
 4. Search for a service.
 5. Find service providers.
 6. View a provider's profile.
 7. See relevant provider information.
 8. Contact the provider.
 9. Hire the provider.
 10. Complete the service process.
 11. Make payment using an available payment method.
 12. Rate and review the provider.
-

Job Marketplace Scenario

A customer must be able to:

1. Create a job.
 2. Enter the required job information.
 3. Publish the job.
 4. Receive applications.
 5. View proposed prices.
 6. View estimated completion times.
 7. Compare applicants.
 8. Select a provider.
 9. Communicate with the selected provider.
 10. Complete the service process.
 11. Make payment.
 12. Submit a review.
-

Worker Scenario

A worker must be able to:

1. Register.
2. Verify their account.
3. Create a profile.
4. Add services.
5. Add experience.
6. Add pricing information.
7. Add availability.

8. Upload portfolio images.
 9. Upload certificates.
 10. View suitable jobs.
 11. Apply for a job.
 12. Submit a proposed price.
 13. Submit estimated completion time.
 14. Communicate with customers.
 15. Complete jobs.
 16. Receive customer ratings.
-

Business Scenario

A business must be able to:

1. Register.
 2. Create a business profile.
 3. Add services.
 4. Provide business information.
 5. Add portfolio information.
 6. Receive customer requests.
 7. Apply for jobs.
 8. Submit proposed prices.
 9. Provide completion estimates.
 10. Communicate with customers.
 11. Complete services.
 12. Receive customer ratings and reviews.
-

40. Conclusion

ServiceHub is designed as a digital marketplace that connects customers with trustworthy individual workers and businesses.

The platform addresses the difficulty customers face when attempting to find reliable service providers while simultaneously providing skilled workers and businesses with a structured way to reach customers.

The platform's central concept is based on **trust, discoverability, and choice**.

Customers can discover providers through search and filtering, examine their professional profiles, review ratings and portfolios, communicate with them, and hire them directly.

At the same time, customers can post their own service requirements and allow workers and businesses to compete for the opportunity by submitting applications containing proposed prices and estimated completion times.

This creates two complementary ways of obtaining services:

Customer-driven discovery

Find a provider → Evaluate → Contact → Hire

and

Provider-driven discovery

Post a job → Receive applications → Compare → Select → Hire

The platform will initially support customers, individual workers, and businesses, with administrators responsible for maintaining and managing the marketplace.

The first version will be available through both web and mobile applications, using Next.js/React for the web frontend, Flutter for mobile, Node.js and Express.js for the backend, and PostgreSQL for data storage.

Payment support will initially include cash, Telebirr, and Chapa, while ServiceHub will generate revenue through a commission taken from workers. The exact commission percentage remains to be determined.

The platform will initially operate in English, with additional languages, location-based functionality, analytics, advanced verification, escrow, and other advanced capabilities reserved for future versions.

Ultimately, ServiceHub aims to provide a more organized, transparent, and trustworthy way for people to find and hire service providers while giving skilled professionals and businesses greater access to customers and opportunities.